ENGS 3302 PATTERN RECOGNITION · STUDENT NOTES

Week 3: Probability for Pattern Recognition

Sources: Duda, Hart & Stork, *Pattern Classification*, 2nd ed., Appendix A.4 (from p. 611) and §2.1 · Bishop, *Pattern Recognition and Machine Learning*, §1.2.1–1.2.4 · Palmer Penguins data: Horst, Hill & Gorman (2020), CC0 · course lecture notes

What you must be able to do after this week (exam scope)

1. Compute probabilities, joint probabilities and conditional probabilities from a table of counts.
2. Use the product rule for independent events.
3. Apply Bayes' rule with given numbers, and explain why $P(A \mid B)$ and $P(B \mid A)$ are different.
4. Write the probability mass function (PMF) of a simple discrete random variable, check that it adds up to 1, and compute its expected value.
5. Turn a histogram into a density, and read a probability as an area under a density.
6. Explain the difference between a probability P and a density p: why $P(X = a) = 0$ for a continuous feature and why a density can be larger than 1.
7. Compute a normal density at a point and the probability of an interval (with `norm.pdf` and `norm.cdf`), use the 68% and 95% rule, and fit a normal curve to data with its mean and standard deviation. (In a paper exam, the values of `norm.cdf` that you need are given.)
8. Explain why comparing two class densities at the same x is a fair comparison.

Not required: the axioms of probability, integrals and the formula of the cumulative distribution, the variance of a random variable, counting formulas (combinations), the multivariate normal (Week 5).

Before you start: read the *Week 3 Math Toolkit* (fractions, the sum sign Σ, the exponential e^x, areas of rectangles) and try the Math check on Moodle.
Reading (optional, these notes are complete): Bishop §1.2.1–1.2.4 (pp. 12–28), on probability, densities and the normal distribution.

3.1 Why probability?

[Duda App. A.4]

In Week 2 the nearest-mean classifier gave a hard answer: *sea bass* or *salmon*. But real classes overlap. Some salmon are light, and some sea bass are dark. So a good system should not only give an answer. It should also say *how likely* each answer is. Probability is the language for this.

An **experiment** is something with an uncertain result: we catch a fish, roll a die, or receive an e-mail. An **event** is a set of results, such as "the fish is a sea bass" or "the die shows 6".

What is the **probability** $P(A)$ of an event A? Repeat the experiment many times. Count how often A happens, and divide by the number of trials. This fraction changes a little after each trial. But after many trials it settles near one value. That value is the probability:

$$P(A) \approx \frac{\text{number of times } A \text{ happens}}{\text{number of trials}}.$$

Figure 1 shows a computer rolling a fair die 10 000 times. After 10 rolls the fraction of sixes is 0.100. After 100 rolls it is 0.160, and after 10 000 rolls it is 0.169. The true probability is $1/6 = 0.167$. The fraction does not get closer at every roll: the line goes up and down. But the swings become smaller as the rolls add up.

This matters for us. In pattern recognition we *estimate* probabilities from training data. A class with only a few samples gives estimates we cannot trust much.

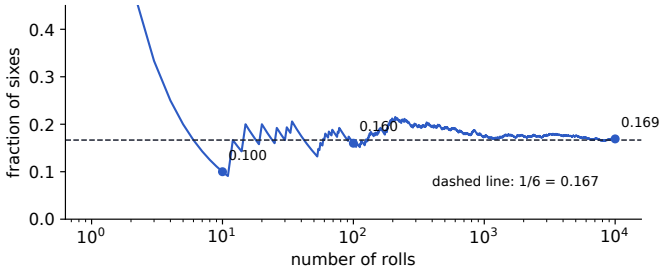


Figure 1: The fraction of sixes after n rolls of a fair die (computer simulation, `random seed 0`).

Basic rules

- $0 \leq P(A) \leq 1$. An impossible event has $P = 0$. A certain event has $P = 1$.
- The probabilities of all possible results add up to 1.
- Complement: $P(\text{not } A) = 1 - P(A)$.

3.2 Joint and conditional probability

[Duda App. A.4]

Table 1 is a day's catch of 100 fish (made-up numbers for practice). We know two things about each fish. First, its class: sea bass or salmon. Second, whether it is light or dark. We call a fish *light* when its lightness is above 5.5. This turns a measurement into a yes/no feature.

	light	**dark**	**total**
sea bass	56	4	60
salmon	3	37	40
total	59	41	100

Table 1: A day's catch: class and lightness of 100 fish.

- A **joint probability** $P(A, B)$ is the probability that A *and* B both happen. Example: $P(\text{sea bass}, \text{light}) = 56/100 = 0.56$.
- A **conditional probability** $P(A \mid B)$, read "A given B", is the probability of A when we already know that B happened. Because we know B happened, we ignore every fish where B did not happen. We look only at the row (or column) of B, and we count inside it.

Conditional probability

$$P(A \mid B) = \frac{P(A, B)}{P(B)} \qquad \Longleftrightarrow \qquad P(A, B) = P(A \mid B)\, P(B) \quad \text{(product rule)}$$

The first formula needs $P(B) > 0$, because we cannot divide by zero.

Worked example 3.1 — reading the catch table

$P(\text{sea bass}) = 60/100 = 0.60$ $P(\text{light}) = 59/100 = 0.59$

$P(\text{light} \mid \text{sea bass}) = 56/60 = \mathbf{0.933}$ (inside the sea bass row)

$P(\text{sea bass} \mid \text{light}) = 56/59 = \mathbf{0.949}$ (inside the light column)

$P(\text{light} \mid \text{salmon}) = 3/40 = 0.075$ $P(\text{salmon} \mid \text{dark}) = 37/41 = 0.902$

Joint or conditional? $P(\text{sea bass}, \text{light}) = 56/100 = 0.56$, but $P(\text{light} \mid \text{sea bass}) = 56/60 = 0.933$. The top number is the same 56 fish. The bottom number is different. The joint probability divides by all 100 fish. The conditional probability divides only by the 60 sea bass. So always ask: *out of which group am I counting?*

Be careful

$P(\text{light} \mid \text{sea bass})$ and $P(\text{sea bass} \mid \text{light})$ are *different questions*. The first counts inside the sea bass row (56 of 60). The second counts inside the light column (56 of 59). Here the two answers are close (0.933 and 0.949), but only because this catch has many sea bass. In Worked example 3.3 sea bass are rare. Then $P(\text{light} \mid \text{sea bass})$ is still 0.933, but $P(\text{sea bass} \mid \text{light})$ falls to 0.396. Never swap the two.

Independence. Two events are **independent** when $P(A, B) = P(A)\, P(B)$. There is an easier way to say this: $P(A \mid B) = P(A)$. In words, learning that B happened does not change the chance of A.

Are lightness and class independent? Test it with numbers. If they were, we would expect $P(\text{sea bass}, \text{light}) = 0.60 \times 0.59 = 0.354$, so about 35 light sea bass in 100 fish. The table has 56. So they are *not* independent, and this is good news. If knowing the lightness did not change the probability of the class, measuring it would not help us classify.

3.3 Bayes' rule: turning a conditional around

[Bishop §1.2, pp. 12–17]

We usually know how each class looks. For example, we know that 93.3% of sea bass are light: that is $P(\text{light} \mid \text{sea bass})$. But a classifier faces the opposite question. It sees a light fish and asks: *is it a sea bass?* That is $P(\text{sea bass} \mid \text{light})$. Bayes' rule turns the first number into the second.

Bayes' rule

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}, \qquad P(B) = P(B \mid A)\, P(A) + P(B \mid \text{not } A)\, P(\text{not } A).$$

The second formula is called **total probability**. A light fish is either a light sea bass or a light salmon. So the probability of "light" is the sum of these two parts.

Worked example 3.2 — why use Bayes' rule if we can count?

On the day of Table 1 we can simply count: $P(\text{sea bass} \mid \text{light}) = 56/59 = 0.949$. Bayes' rule gives the same answer from other numbers: $P(\text{light}) = 0.933 \times 0.60 + 0.075 \times 0.40 = 0.56 + 0.03 = 0.59$ (light sea bass plus light salmon), and $P(\text{sea bass} \mid \text{light}) = 0.56/0.59 = \mathbf{0.949}$.
The agreement is not the lesson. The lesson is *which numbers* Bayes' rule needs:

- how sea bass look: $P(\text{light} \mid \text{sea bass}) = 0.933$;
- how salmon look: $P(\text{light} \mid \text{salmon}) = 0.075$;
- how common each class is: 0.60 and 0.40.

The first two describe the fish themselves, so we assume they stay the same from day to day. The third changes with the season, the place and the net. So we measure the first two once. On a new day we only need the new class shares, not a new table. Worked example 3.3 does exactly this.

Worked example 3.3 — when sea bass are rare

Another day, only 5% of the catch is sea bass. The fish themselves have not changed: 93.3% of sea bass are still light, and 7.5% of salmon are still light.
Think of 1000 fish. 50 are sea bass, and about $50 \times 0.933 \approx 47$ of them are light. 950 are salmon, and about $950 \times 0.075 \approx 71$ of them are light. So there are about $47 + 71 = 118$ light fish, and only 47 of them are sea bass: $47/118 \approx 0.40$.
Bayes' rule gives the same answer without imagining 1000 fish:

$$P(\text{sea bass} \mid \text{light}) = \frac{0.933 \times 0.05}{0.933 \times 0.05 + 0.075 \times 0.95} = \frac{0.0467}{0.0467 + 0.0713} = \mathbf{0.396}.$$

A light fish is now *more likely a salmon*, even though light fish are typical of sea bass. Why? Each salmon has only a small chance of being light. But there are so many salmon that they still make most of the light fish. With 60% sea bass the answer was 0.949. With 20% it is 0.757. With 5% it is 0.396. The rate inside each class did not change. Only the share of each class changed, and that was enough. Week 4 calls the class share the *prior* and the answer the *posterior*.

3.4 Discrete random variables and the PMF

[Duda App. A.4.1]

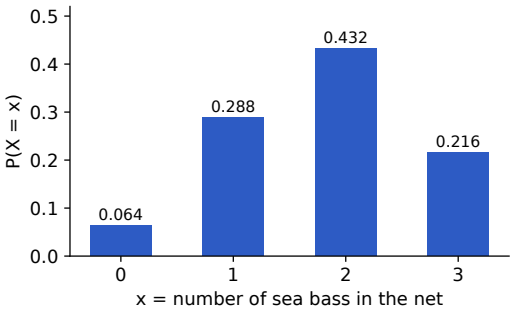
A **random variable** X is a number that we get from an experiment. For example, X can be the number of sea bass in a net. X is called **discrete** when it can take only separate values, like $0, 1, 2, 3$. It cannot be 1.5. Counts are always discrete. The **probability mass function** (PMF) of X gives the probability of each value: $P(X = x)$.

Worked example 3.4 — sea bass in a net of three fish

Each fish in the net is a sea bass with probability 0.6, independently of the others. Let X be the number of sea bass. One order of results, sea bass–sea bass–salmon, has probability $0.6 \times 0.6 \times 0.4 = 0.144$ (product rule). But two sea bass can come in three orders (BBS, BSB, SBB). So $P(X = 2) = 3 \times 0.144 = 0.432$. In the same way:

x	0	1	2	3
$P(X = x)$	0.064	0.288	0.432	0.216

Check: $0.064 + 0.288 + 0.432 + 0.216 = 1$.



This check is not a formality. A common mistake is to forget that the sea bass can come in different orders. Then the four numbers are 0.064, 0.096, 0.144 and 0.216, and they add up to only 0.52. When a PMF does not add up to 1, something is missing.

PMF and expected value

$$P(X = x) \geq 0, \qquad \sum_x P(X = x) = 1, \qquad \text{expected value } E[X] = \sum_x x\, P(X = x).$$

The **expected value** is the long-run average of X. For the net: $E[X] = 0(0.064) + 1(0.288) + 2(0.432) + 3(0.216) = 1.8$ sea bass per net. This equals 3×0.6: three fish, each a sea bass with probability 0.6. Notice that 1.8 is not a possible value: no net holds 1.8 sea bass. The expected value is the average over many nets, not a prediction for one net. (The most likely single value is 2, with probability 0.432.)

3.5 Continuous features: from histogram to density

[Bishop §1.2.1, p. 17]

Lightness, width and flipper length are **continuous**: they can take any value in a range. We cannot list a probability for every value, so we describe them with a **density**.

A **density** is a count divided by a size, like the number of people per km^2. For a feature like flipper length, the size is a length in mm. Figure 2(a) is a histogram of the flipper length of 342 penguins, with 5 mm bins. We change it in two steps. **Step 1:** divide each count by the bin width (5 mm). Now each bar shows *penguins per mm*: how crowded the penguins are in that bin. We call this the **count density**. **Step 2:** divide by all 342 penguins. Now the height is the *fraction* of the penguins per mm (Figure 2(b)). We call this the **probability density**, or simply the density. It does not depend on how many penguins we measured, and it is the $p(x)$ we use from now on. To get back to a fraction of penguins, multiply the height by the width. So the *area* of each bar is the fraction of penguins in that bin, and all the areas add up to 1.

Worked example 3.5 — density means "how many per mm"

Flipper lengths are recorded in whole mm. 62 penguins have a length from 190 to 194 mm, a range 5 mm wide.
Count density. $62/5 = \mathbf{12.4}$ penguins per mm, on average. They are not spread evenly: 22 penguins have 190 mm, but only 7 have 192 mm. The bar shows the average.
Probability density. Divide by all 342 penguins: $12.4/342 = \mathbf{0.0363}$ per mm. This is the height of the bar: on average, 3.6% of the penguins fall in each mm of this range. Each step here is 1 mm, so 0.0363 is also the probability of one mm, on average. This works only because the step is 1 mm: per metre the density is 36.3, but one mm still has probability 0.0363.

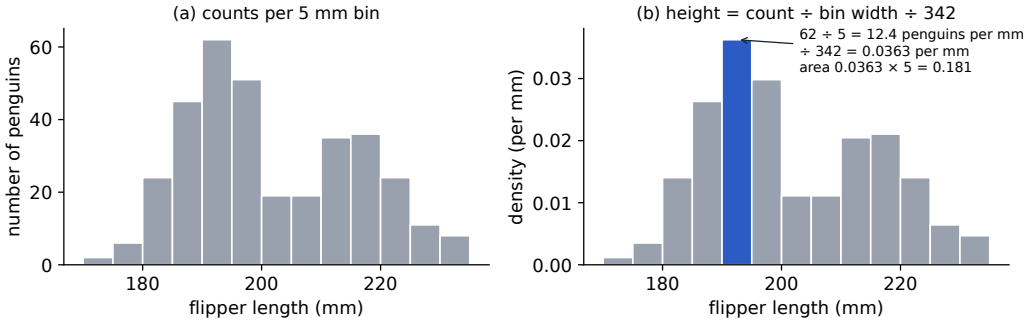


Figure 2: Flipper length of 342 penguins. (a) Counts. (b) Density: the area of each bar is the fraction of penguins in it.

Area. $0.0363 \times 5 = 0.181 = 62/342$: the fraction of penguins in the bar.
A wider bar. From 190 to 199 mm there are 113 penguins in 10 mm: 11.3 per mm, so $11.3/342 = 0.0330$ per mm. This bar holds more penguins (0.330 against 0.181), but it is *lower*, because they are less crowded. The height tells us how crowded the penguins are. Only the area tells us how many. So a height is not a probability; an area is.
Try it: Density Explorer, Part 1. Compare two bars: the taller one can hold fewer penguins. Then switch between count density and probability density, and between mm and m.

With more data and narrower bins, the bars get closer and closer to a smooth curve. This curve is the **probability density function** (PDF) $p(x)$.

A probability density function $p(x)$

- $p(x) \geq 0$, and the total area under the curve is 1.
- The probability of an interval is the **area** under the curve over that interval: $P(a < X < b) =$ area from a to b.
- The probability of one exact value is zero: $P(X = a) = 0$. A single point has no width, so it has no area. This does not mean the value is impossible: every fish has some exact lightness. So for a continuous feature we never ask "what is the probability of exactly 6?". We ask about an interval, such as 5.5 to 6.5. A single point adds no area, so it does not matter whether we write $<$ or $\leq$.

Probability P or density p? (Duda §2.1)

We write a capital P for a probability and a small p for a density. A density is *not* a probability. It is "probability per unit of x", so a density can be larger than 1. In Figure 3, the same sea bass lightness curve has peak 0.399 on a 0–10 scale and peak 3.99 on a 0–1 scale. Both curves have area 1. Why? On the 0–1 scale the curve is 10 times narrower, so it must be 10 times taller to keep the area equal to 1. The probabilities do not change: $P(6 < X < 8) = 0.683$ on the 0–10 scale, and $P(0.6 < X < 0.8) = 0.683$ on the 0–1 scale. Real data does the same. In Worked example 3.5 there are 12.4 penguins per mm, which is 12 400 penguins per metre: the same penguins, a different unit. In the same way, the Gentoo flipper curve of Section 3.6 has peak 0.062 per mm, or 61.8 per metre. A density can be any positive number. A probability can never be larger than 1.

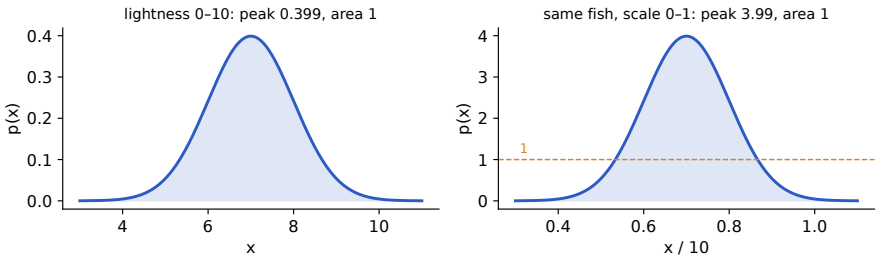


Figure 3: The same density on two scales. Squeezing the x-axis by 10 makes the curve 10 times taller.

3.6 The normal density

[Bishop §1.2.4, p. 24]

Many measurements give a bell-shaped histogram. The **normal** (Gaussian) density describes it with only two numbers: the mean μ, which is the centre, and the standard deviation σ, which is the spread.

Normal density $N(\mu, \sigma^2)$ (Duda App. A.4.12, Eqs. 91–93)

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) = \frac{0.3989}{\sigma}\, e^{-z^2/2}, \qquad z = \frac{x-\mu}{\sigma}.$$

Be careful with the second number. In $N(\mu, \sigma^2)$ it is the variance σ^2. But Python's `norm.pdf(x, mu, sigma)` wants the standard deviation σ. So for $N(7, 4)$ you write `norm.pdf(x, 7, 2)`. About 68% of the area lies within $\mu \pm \sigma$, and about 95% within $\mu \pm 2\sigma$.

Worked example 3.6 — a fish with lightness 6

The models of Week 4: sea bass lightness $N(7, 1)$, salmon lightness $N(4, 1)$.
Sea bass: $z = (6-7)/1 = -1$, $\quad p(6) = 0.3989 \times e^{-0.5} = 0.3989 \times 0.6065 = \mathbf{0.242}$.
Salmon: $z = (6-4)/1 = 2$, $\quad p(6) = 0.3989 \times e^{-2} = 0.3989 \times 0.1353 = \mathbf{0.054}$.
Why is the sea bass density higher? The value 6 is 1 standard deviation from the sea bass centre, but 2 standard deviations from the salmon centre. The further x is from the centre, counted in standard deviations, the lower the curve.
At $x = 6$ the sea bass density is about $0.242/0.054 \approx 4.5$ times the salmon density. In words: a lightness of 6 fits the sea bass model much better than the salmon model. This is not yet the probability that the fish is a sea bass. For that we also need how common each class is (Worked example 3.3, and Week 4).

The probability of an interval needs an area. In Python, `norm.cdf(b, mu, sd)` gives the area to the left of b. So the area between a and b is `norm.cdf(b)` − `norm.cdf(a)`. For sea bass, $P(6 < X < 8) = 0.841 - 0.159 = 0.683$ (Figure 4), as the 68% rule says.

`norm.cdf` also shows where Table 1 came from. "Light" means lightness above 5.5. For sea bass, $P(X > 5.5) = 1 -$ `norm.cdf(5.5, 7, 1)` $= 1 - 0.067 = 0.933$. This is the 56 of 60 we counted in the sea bass row: the made-up table was built from this model. (For salmon the model gives 0.067. The table has $3/40 = 0.075$, because $40 \times 0.067 = 2.7$ was rounded to 3 fish.)

Fitting a normal curve to data. To fit a normal curve, compute two numbers from the data: the mean and the standard deviation. Use them as μ and σ. (Is this the best choice? Yes. In Week 5 we will see why.) For the 123 Gentoo penguins: mean 217.2 mm, standard deviation 6.46 mm. Figure 5

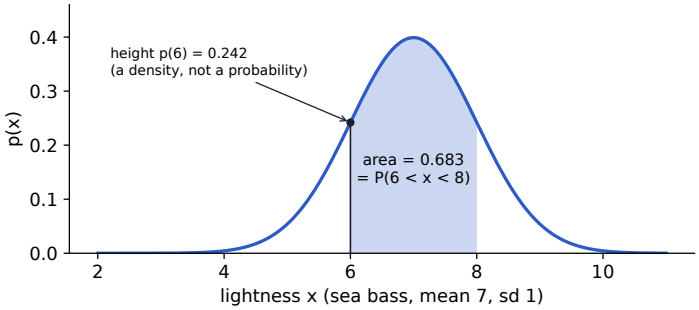


Figure 4: The height at 6 is a density; the shaded area from 6 to 8 is a probability.

draws one fitted curve per species.

The curve is a model, not the data, so let us compare them. Flipper lengths are rounded to whole millimetres. A penguin recorded as 210 mm really has a length somewhere between 209.5 and 210.5 mm. So "210 to 220 mm" in the data means the interval from 209.5 to 220.5 mm under the curve. The model gives $P(209.5 < X < 220.5) = 0.579$, and 60.2% of the Gentoo samples really are in this range. The two numbers are close. So a curve with only two numbers describes 123 penguins well, though not perfectly.

A normal curve fits well only when the histogram has one hump. Suppose we fit one normal curve to all 342 penguins together (mean 200.9 mm, standard deviation 14.0 mm). Its peak sits near 201 mm, where the real data has a dip. The curve predicts about 92 penguins from 200 to 209 mm, but only 38 are really there. This is why we fit one curve per class: the class-conditional density $p(x \mid \omega)$ of Section 3.7.

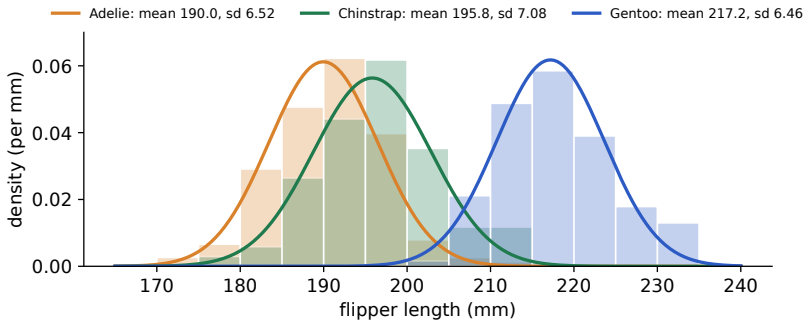


Figure 5: Density histograms and fitted normal curves for the three species. Adelie and Chinstrap overlap strongly.

3.7 Why we can compare densities

[Duda §2.1, p. 21 · Bishop §1.2.1, p. 17]

A density is not a probability. So is it fair to compare $p(6 \mid \text{sea bass}) = 0.242$ with $p(6 \mid \text{salmon}) = 0.054$? Yes. Take a thin strip of width Δ around x. Under the curve, the strip is almost a rectangle, so its probability is about $p(x)\,\Delta$ (Figure 6).

Worked example 3.7 — strips of different widths around $x = 6$

strip	sea bass P	salmon P	ratio
width 1 (5.5 to 6.5)	0.242	0.061	4.0
width 0.1 (5.95 to 6.05)	0.0242	0.0054	4.48
width 0.01 (5.995 to 6.005)	0.00242	0.00054	4.48

The probabilities depend on the width we choose: a strip 10 times thinner holds 10 times less probability. But once the strip is thin, the ratio does not depend on the width. It settles at 4.48, which is 0.242/0.054, the ratio of the two densities. So comparing densities is the same as comparing the probabilities of equal thin strips. For the wide strip (width 1) the rectangle picture is poor, and the ratio is off (4.0).

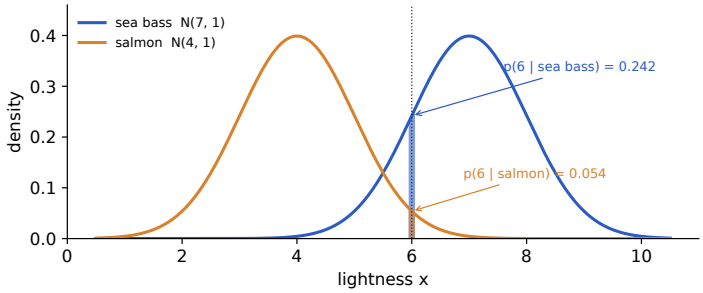


Figure 6: The two class densities at $x = 6$. A thin strip of the same width under each curve turns the heights into probabilities.

The notation $p(x \mid \omega)$ means the density of x for the fish of class ω. It is called a **class-conditional density**. Week 4 puts the two ideas of this week together. First, compare the class densities at the measured x. Second, take into account how common each class is, as in Worked example 3.3. Together, these two steps are the Bayes decision.

3.8 In Python

```
import numpy as np
from scipy.stats import norm
rng = np.random.default_rng(0)
rolls = rng.integers(1, 7, 10000)        # 10 000 rolls of a die
print(np.mean(rolls == 6))               # 0.1691, close to 1/6

print(56 / 59)                           # P(sea bass | light), counted: 0.949
print(0.933*0.05 / (0.933*0.05 + 0.075*0.95))  # rare day: 0.396

print(norm.pdf(6, 7, 1), norm.pdf(6, 4, 1))    # densities: 0.242, 0.054
print(norm.cdf(8, 7, 1) - norm.cdf(6, 7, 1))   # probability of an interval: 0.683
```

A density histogram (bar areas add up to 1): `plt.hist(x, bins=20, density=True)`. The numbers for a fitted normal curve: `np.mean(x)` and `np.std(x)`. (`np.std` divides by n. pandas `.std()` divides by $n - 1$ and gives 6.48 for Gentoo.)

Summary

- A probability is a long-run fraction: $0 \le P \le 1$, and all results add up to 1.
- Conditional probability: $P(A \mid B) = P(A, B)/P(B)$. Count inside the row or column of B. Independent events: $P(A, B) = P(A)P(B)$.
- Bayes' rule turns $P(B \mid A)$ into $P(A \mid B)$. The share of each class can change the answer a lot.
- A discrete random variable has a PMF $P(X = x)$ that adds up to 1. Its expected value is $\sum x\, P(X = x)$, the average over many trials.
- A continuous feature has a density $p(x)$. Probability is area, $P(X = a) = 0$, and $p(x)$ can be larger than 1.
- The normal density $N(\mu, \sigma^2)$ has mean μ and variance σ^2. `norm.pdf` gives a height, `norm.cdf` an area.
- Densities of two classes at the same x can be compared, because equal thin strips turn them into probabilities.

Check yourself

1. From Table 1, compute three numbers about the 3 light salmon: $P(\text{salmon}, \text{light})$, $P(\text{salmon} \mid \text{light})$ and $P(\text{light} \mid \text{salmon})$. All three have 3 on top. Why are they different?
2. A machine alarm rings for 90% of faulty parts and for 10% of good parts; 2% of parts are faulty. (a) Compute $P(\text{alarm})$. (b) Compute $P(\text{faulty} \mid \text{alarm})$. (c) Why is (b) so much smaller than 0.9?
3. The number of links X in an e-mail has the PMF $P(0) = 0.5$, $P(1) = 0.3$, $P(2) = 0.15$, $P(3) = 0.05$. Compute $E[X]$ and $P(X \ge 2)$. Can one e-mail have $E[X]$ links? What does $E[X]$ mean?
4. Two bars from density histograms of the 342 flipper lengths. Bar A covers 195 to 199 mm (5 mm wide) and has height 0.0298 per mm. Bar B covers 180 to 189 mm (10 mm wide) and has height 0.0202 per mm. (a) Which bar is taller? (b) What fraction of the penguins is in each bar? (c) Which interval holds more penguins?
5. Sea bass lightness is $N(7, 1)$. Find $p(8)$, $P(X < 6)$ (use `norm.cdf`) and $P(X = 6)$. Why is $p(8)$ equal to $p(6)$? Why is $P(X = 6) = 0$ while $p(6) = 0.242$ is not zero?
6. A fish has lightness 3. Compute $p(3 \mid \text{salmon})$ and $p(3 \mid \text{sea bass})$ with the models of Worked example 3.6 ($e^{-0.5} = 0.6065$, $e^{-8} = 0.000335$). How many times larger is the salmon density? Why is it fair to compare these two numbers, even though neither is a probability?

Answers: 1. $3/100 = 0.03$ (out of all fish); $3/59 = 0.051$ (out of the light fish); $3/40 = 0.075$ (out of the salmon). 2. (a) $0.9(0.02) + 0.1(0.98) = 0.116$; (b) $0.018/0.116 = 0.155$; (c) in 1000 parts, 20 are faulty and give 18 alarms, 980 are good and give 98 alarms: good parts are so many that their false alarms outnumber the true ones. 3. 0.75; 0.20. No: 0.75 is the average number of links over many e-mails. 4. (a) A. (b) A: $0.0298 \times 5 = 0.149$ (51 penguins); B: $0.0202 \times 10 = 0.202$ (69 penguins). (c) B: the lower bar holds more penguins. 5. 0.242; 0.159; 0. Both 6 and 8 are 1 standard deviation from 7, and the curve is symmetric. One point has no width, so no area; $p(6)$ is a height. 6. 0.242 and 0.000134; about 1800 times. Thin strips of the same width around 3 turn both densities into probabilities, and the ratio stays the same.